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**Instructions** Follow instructions *carefully*, failure to do so will result in points being deducted. You may write or type out your answers. Be sure to show sufficient work to justify your answer(s).

1. 6 points Use set builder notation to give a description of each of these sets

(a)  $\{-15, -10, -5, 0, 5, 10, 15\}$

(b)  $\{0, 6, 12, \dots, 66\}$

**Solution:** Enter your solution here:

A:

$$\{x \mid (x = 5k), k \in \mathbb{Z}, -3 \leq k \leq 3\}$$

B:

$$\{x \mid (x = 6k), k \in \mathbb{Z}, 0 \leq k \leq 11\}$$

2. 6 points Determine whether these statements are true or false.

(a)  $2026 \in \{2026\}$

(b)  $\{2026\} \in \{2026\}$

(c)  $\{2026\} \subseteq \{2026\}$

(d)  $\{2026\} \subseteq \{\{2026\}\}$

(e)  $\{2026\} \subsetneq \{2026, \{2026\}\}$

(f)  $\{\{2026\}\} \subsetneq \{\{2026\}, \{2026\}\}$

**Solution:** Enter your solution here:

A:

$$2026 \in \{2026\} \quad \text{True}$$

B:

$$\{2026\} \in \{2026\} \quad \text{False}$$

C:

$$\{2026\} \subseteq \{2026\} \quad \text{True}$$

D:

$$\{2026\} \subseteq \{\{2026\}\} \quad \text{False}$$

E:

$$\{2026\} \subsetneq \{2026, \{2026\}\} \quad \text{True}$$

F:

$$\{\{2026\}\} \subsetneq \{\{2026\}, \{2026\}\} \quad \text{False}$$

3. 6 points What is the cardinality of each of the following?

- (a) 2026
- (b)  $\{\}$
- (c)  $\{2026\}$
- (d)  $\{2026, 2026\}$
- (e)  $\{2026, \{2026\}\}$
- (f)  $\{2026, \{2026\}, \{2026, \{2026\}\}\}$

**Solution:** Enter your solution here:

A: Undefined - Not a set

B: 0

C: 1

D: 1

E: 2

F: 3

4. 8 points Let  $A = \{\text{hydrogen, oxygen, sodium}\}$ ,  $B = \{\text{sodium, zinc}\}$ , and  $C = \{\text{hydrogen, boron, lead}\}$ . Find the following

- (a)  $P(A)$
- (b)  $P(A \cup B)$
- (c)  $A \times C$
- (d)  $A \times B \times C$

**Solution:** Enter your solution here:

Given:

$A = \{\text{H, O, Na}\}$

$B = \{\text{Na, Zn}\}$

$C = \{\text{H, Bo, Pb}\}$

Where: H = hydrogen, O = oxygen, Na = sodium, Zn = zinc, Bo = boron, Pb = lead

A:

$$A = \{H, O, Na\}$$

$$P(A) = \{\emptyset, \{H\}, \{O\}, \{Na\}, \{H, O\}, \{H, Na\}, \{O, Na\}, \{H, O, Na\}\}$$

B:

$$A \cup B = \{H, O, Na, Zn\}$$

$$P(A \cup B) = \{\emptyset, \{H\}, \{O\}, \{Na\}, \{Zn\}, \{H, O\}, \{H, Na\}, \{H, Zn\}, \{O, Na\}, \{O, Zn\}, \{Na, Zn\}, \{H, O, Na\}, \{H, O, Zn\}, \{H, Na, Zn\}, \{O, Na, Zn\}, \{H, O, Na, Zn\}\}$$

C:

$$A = \{H, O, Na\}$$

$$C = \{H, Bo, Pb\}$$

$$A \times C = \{(H, H), (H, Bo), (H, Pb), (O, H), (O, Bo), (O, Pb), (Na, H), (Na, Bo), (Na, Pb)\}$$

D:

$$A = \{H, O, Na\}$$

$$B = \{Na, Zn\}$$

$$C = \{H, Bo, Pb\}$$

$$A \times B \times C = \{(H, Na, H), (H, Na, Bo), (H, Na, Pb), (H, Zn, H), (H, Zn, Bo), (H, Zn, Pb), (O, Na, H), (O, Na, Bo), (O, Na, Pb), (O, Zn, H), (O, Zn, Bo), (O, Zn, Pb), (Na, Na, H), (Na, Na, Bo), (Na, Na, Pb), (Na, Zn, H), (Na, Zn, Bo), (Na, Zn, Pb)\}$$

5. 8 points Let A be the set of students who attend the Capybara cafe, and let B be the set of students who drink Shamrock Shakes. Describe the students in each of these sets.

- (a)  $A \cap B$
- (b)  $A \cup B$
- (c)  $A - B$
- (d)  $B - A$

**Solution:** Enter your solution here:

A:  $A \cap B$

Students who attend the Capybara cafe and drink Shamrock Shakes.

B:  $A \cup B$

Students who attend the Capybara cafe or drink Shamrock Shakes.

C:  $A - B$

Students who attend the Capybara cafe and do not drink Shamrock Shakes.

D:  $B - A$

Students who drink Shamrock Shakes and do not attend the Capybara cafe.

6. 6 points Find the sets A and B if  $A - B = \{\text{apple, grape, banana, mango}\}$ ,  $B - A = \{\text{blackberry, raspberry}\}$ , and  $A \cap B = \{\text{strawberry, kiwi, pineapple}\}$ .

**Solution:** Enter your solution here:

A: {apple, grape, banana, mango, strawberry, kiwi, pineapple}

B: {blackberry, raspberry, strawberry, kiwi, pineapple }

7. 6 points What can you say about the sets A and B if we know that

(a)  $A \cap B = B$

(b)  $A \cup B = A$

(c)  $A - B = B - A$

**Solution:** Enter your solution here:

A: B is a subset of A

B: B is a subset of A

C: A and B are equal sets

8. 20 points State whether each of the following is a function, if so prove it is onto, and/or one-to-one for  $f : \mathbb{Z} \rightarrow \mathbb{Z}$ :

(a)  $f(x) = 2x - 3$

(b)  $f(x) = 5x^2$

(c)  $f(x) = -2x^2 + 3$

(d)  $f(x) = \frac{\sqrt{x}}{5}$

**Solution:** Enter your solution here:

**A:**

$f : \mathbb{Z} \rightarrow \mathbb{Z} \quad f(x) = 2x - 3$

**Function:** Yes, this is a valid function since every integer input produces exactly one integer output.

**1-to-1:**

Suppose  $f(x_1) = f(x_2)$ :

$$2x_1 - 3 = 2x_2 - 3$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

Therefore  $f$  would be 1-to-1.

**Onto:** Let  $y = 2x - 3$ , solve for  $x$ :

$$2x = y + 3$$

$$x = \frac{y + 3}{2}$$

If  $y = 0$ , then  $x = \frac{3}{2}$ , which is not an integer, so  $f(x)$  would not be onto.

**B:**

$$f : \mathbb{Z} \rightarrow \mathbb{Z} \quad f(x) = 5x^2$$

**Function:** Yes, this is a valid function since every integer input produces exactly one integer output.

**1-to-1:**

$f$  is **not** 1-to-1. Counterexample:  $f(1) = 5(1)^2 = 5$  and  $f(-1) = 5(-1)^2 = 5$ , so  $f(1) = f(-1)$  therefore  $f$  can not be 1-to-1.

**Onto:**

$f$  is **not onto**. Since  $5x^2 \geq 0$  for all  $x \in \mathbb{Z}$ , no integer  $x$  satisfies  $f(x) = -1$ , so  $f$  can not be onto.

**C:**

$$f : \mathbb{Z} \rightarrow \mathbb{Z} \quad f(x) = -2x^2 + 3$$

**Function:** Yes, this is a valid function since every integer input produces exactly one integer output.

**1-to-1:**

$f$  is **not** 1-to-1. Counterexample:  $f(1) = -2(1)^2 + 3 = 1$  and  $f(-1) = -2(-1)^2 + 3 = 1$ , so  $f(1) = f(-1)$  therefore  $f$  can not be 1-to-1.

**Onto:**

$f$  is **not onto**. Since  $-2x^2 \leq 0$ , we have  $f(x) = -2x^2 + 3 \leq 3$  for all  $x \in \mathbb{Z}$ . Thus no integer  $x$  satisfies  $f(x) = 4$ , so  $f$  can not be onto.

**D:**

$$f : \mathbb{Z} \rightarrow \mathbb{Z} \quad f(x) = \frac{\sqrt{x}}{5}$$

This is **not a valid function** from  $\mathbb{Z} \rightarrow \mathbb{Z}$ .

For example,  $x = -1 \in \mathbb{Z}$  gives  $f(-1) = \frac{\sqrt{-1}}{5}$ , which is not a real number. Since the expression is undefined for negative integers,  $f$  is not defined on all of  $\mathbb{Z}$  and therefore is not a function from  $\mathbb{Z} \rightarrow \mathbb{Z}$ .

9. 20 points State whether each of the following is a function, if so prove it is onto, and/or one-to-one for  $f : \mathbb{R} \rightarrow \mathbb{R}$ :

(a)  $f(x) = 2x - 3$

(b)  $f(x) = 5x^2$

(c)  $f(x) = -2x^2 + 3$

(d)  $f(x) = \frac{\sqrt{x}}{5}$

**Solution:** Enter your solution here:

**A:**

**Function:** Yes, this is a valid function since every real number input produces exactly one real number output.

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = 2x - 3$$

**1-to-1:**

Suppose  $f(x_1) = f(x_2)$ :

$$2x_1 - 3 = 2x_2 - 3$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

Therefore  $f$  would be 1-to-1.

**Onto:** Let  $y = 2x - 3$ , solve for  $x$ :

$$2x = y + 3$$

$$x = \frac{y + 3}{2}$$

If  $y = 0$ , then  $x = \frac{3}{2}$ , which is a real number, so  $f(x)$  would be onto.

**B:**

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = 5x^2$$

**Function:** Yes, this is a valid function since every real number input produces exactly one real number output.

**1-to-1:**

$f$  is **not** 1-to-1. Counterexample:  $f(1) = 5(1)^2 = 5$  and  $f(-1) = 5(-1)^2 = 5$ , so  $f(1) = f(-1)$  therefore  $f$  can not be 1-to-1.

**Onto:**

$f$  is **not onto**. Since  $5x^2 \geq 0$  for all  $x \in \mathbb{R}$ , we have  $f(x) \geq 0$  for all  $x \in \mathbb{R}$ . Therefore, no real number  $x$  satisfies  $f(x) = -1$ , so  $f$  can not be onto.

**C:**

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = -2x^2 + 3$$

**Function:** Yes, this is a valid function since every real number input produces exactly one real number output.

**1-to-1:**

$f$  is **not** 1-to-1. Counterexample:  $f(1) = -2(1)^2 + 3 = 1$  and  $f(-1) = -2(-1)^2 + 3 = 1$ , so  $f(1) = f(-1)$  therefore  $f$  can not be 1-to-1.

**Onto:**

$f$  is **not onto**. Since  $-2x^2 \leq 0$  for all  $x \in \mathbb{R}$ , we have  $f(x) = -2x^2 + 3 \leq 3$  for all  $x \in \mathbb{R}$ . The maximum value of  $f$  is 3 (achieved when  $x = 0$ ). Therefore, no real number  $x$  satisfies  $f(x) = 4$ , so  $f$  can not be onto.

**D:**

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = \frac{\sqrt{x}}{5}$$

This is **not a valid function** from  $\mathbb{R} \rightarrow \mathbb{R}$ .

For negative real numbers like  $x = -1 \in \mathbb{R}$ , we have  $f(-1) = \frac{\sqrt{-1}}{5}$ , which is not a real number. Since the square root function is only defined for non-negative real numbers,  $f$  is not defined on all of  $\mathbb{R}$ . Therefore, this is not a function from  $\mathbb{R} \rightarrow \mathbb{R}$ .

10. 8 points Define the following functions, (assume that the domains/codomains are defined such that each composition is valid) where:

$$\begin{aligned} f(x) &= 2x, \\ g(x) &= \frac{x}{(3+x)}, \\ h(x) &= 5x^2 \end{aligned}$$

- (a)  $f \circ g$   
 (b)  $h \circ g$   
 (c)  $f \circ g \circ h$   
 (d)  $h \circ g \circ f$

**Solution:**

Given:  $f(x) = 2x$ ,  $g(x) = \frac{x}{3+x}$ ,  $h(x) = 5x^2$

(a)  $f \circ g(x) = f(g(x))$ :

$$f\left(\frac{x}{3+x}\right) = 2\left(\frac{x}{3+x}\right) = \boxed{\frac{2x}{3+x}}$$

(b)  $h \circ g(x) = h(g(x))$ :

$$h\left(\frac{x}{3+x}\right) = 5\left(\frac{x}{3+x}\right)^2 = \boxed{\frac{5x^2}{(3+x)^2}}$$

(c)  $f \circ g \circ h(x) = f(g(h(x)))$ :

$$g(h(x)) = g(5x^2) = \frac{5x^2}{3+5x^2}$$

$$f(g(h(x))) = f\left(\frac{5x^2}{3+5x^2}\right) = 2 \cdot \frac{5x^2}{3+5x^2} = \boxed{\frac{10x^2}{3+5x^2}}$$

(d)  $h \circ g \circ f(x) = h(g(f(x)))$ :

$$g(f(x)) = g(2x) = \frac{2x}{3+2x}$$

$$h(g(f(x))) = h\left(\frac{2x}{3+2x}\right) = 5\left(\frac{2x}{3+2x}\right)^2 = \boxed{\frac{20x^2}{(3+2x)^2}}$$

11. 6 points Let  $f(x)$  and  $g(x)$  be two linear functions. Note: a function is *linear* if there exists  $a, b \in \mathbb{R}$  such that  $f(x) = ax + b$ .

- (a) Prove or disprove:  $f \circ g = g \circ f$ .  
(b) Prove or disprove:  $f \circ g$  and  $g \circ f$  are linear functions.

**Solution:**

**A:** Prove or disprove:  $f \circ g = g \circ f$

Disproof by counterexample: Let  $f(x) = 2x + 1$  and  $g(x) = x + 3$ .

Then:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(x + 3) = 2(x + 3) + 1 = 2x + 7 \\(g \circ f)(x) &= g(f(x)) = g(2x + 1) = (2x + 1) + 3 = 2x + 4\end{aligned}$$

Since  $2x + 7 \neq 2x + 4$ , we have  $f \circ g \neq g \circ f$ .

Therefore, the statement is false.

**B:** Prove or disprove:  $f \circ g$  and  $g \circ f$  are linear functions

**Proof:**

Let  $f(x) = a_1x + b_1$  and  $g(x) = a_2x + b_2$  where  $a_1, a_2, b_1, b_2 \in \mathbb{R}$ .

**For  $f \circ g$ :**

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\&= f(a_2x + b_2) \\&= a_1(a_2x + b_2) + b_1 \\&= a_1a_2x + a_1b_2 + b_1\end{aligned}$$

This has the form  $cx + d$  where  $c = a_1a_2$  and  $d = a_1b_2 + b_1$ .

Since  $c, d \in \mathbb{R}$ ,  $(f \circ g)(x)$  is a linear function.

**For  $g \circ f$ :**

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\&= g(a_1x + b_1) \\&= a_2(a_1x + b_1) + b_2 \\&= a_1a_2x + a_2b_1 + b_2\end{aligned}$$

This has the form  $cx + d$  where  $c = a_1a_2$  and  $d = a_2b_1 + b_2$ .

Since  $c, d \in \mathbb{R}$ ,  $(g \circ f)(x)$  is a linear function.

Therefore, both  $f \circ g$  and  $g \circ f$  are linear functions.